

Biresb Biswas

Chennai, India | +91-7058844518 | contactbiresb@gmail.com | [LinkedIn](#) ↗ | [GitHub](#) ↗ | [Portfolio](#) ↗

SUMMARY

Software engineer who designs and ships **production systems** across distributed infrastructure, AI, and blockchain. Owns the full engineering lifecycle, from architecture and API/protocol design through implementation, testing, deployment, and operational reliability. Known for turning ambiguous requirements into **maintainable platforms**, developer tooling, and measurable product outcomes.

EXPERIENCE

Summer of Bitcoin — Software Engineer

May 2026 – Present

Rust · Distributed Systems · Bitcoin/Lightning · REST/gRPC · Protobuf · SQLite · AWS EC2

- **Re-architected a monolithic payment backend** into two independently deployable Rust daemons, delivering 26 authenticated REST endpoints over an internal gRPC/Protobuf service layer.
- **Designed and evolved a secure application-layer protocol** using signed and versioned messages, request correlation, resource binding, replay protection, deduplication, and backward-compatible protocol upgrades.
- **Built and deployed crash-safe financial workflows on AWS EC2** using atomic SQLite transactions, idempotent processing, persistent retries, restart recovery, and audit trails across two publicly accessible daemon processes.
- **Contributed application-defined payment metadata support to the upstream Lightning Dev Kit (LDK) Server** across Protobuf APIs, core conversions, CLI tooling, and end-to-end tests.

Nethermind — Software Engineer Intern

Jan 2026 – Apr 2026

DeFi · Multi-Chain · Agentic Systems · Python

- Led **0 to 1 research and product development**: benchmarked **5 LLMs on 400 cases derived from 200 decoded mainnet transactions** and iterated on simulation and agent-interface prototypes, converging on a split between probabilistic intent extraction and deterministic transaction construction.
- **Architected and shipped [defi-skills](#)** ↗, a data-driven engine producing unsigned, auditable transactions for **54 actions across 14 protocol playbooks and 6 EVM chains**, enabled standard integrations without core-engine changes.
- **Owned the product roadmap and end-to-end delivery**, spanning the Python engine, FastAPI service, CLI/PyPI distribution, **Claude and OpenClaw** integrations, browser playground, authentication, rate limiting, CI, and deployment.
- Led the open-source launch, established contribution and validation workflows and reviewed external PRs, growing the project to **7 contributors and 200+ PyPI downloads in its launch week**.

Samsung R&D — Research Intern (Part-time)

Sep 2025 to Apr 2026

Distributed Training · PyTorch · DeepSpeed · Hugging Face PEFT

- Designed **SplitLoRA**, a distributed fine-tuning architecture that combines layer-wise model partitioning with LoRA adapter allocation across heterogeneous CPU, GPU, and multi-device topologies.
- Implemented DeepSpeed/FSDP-based sharding for constrained devices and built benchmarks to quantify scalability, throughput, memory, and downstream-accuracy trade-offs.

PROJECTS

Outreach: AI-powered lead-gen platform | *FastAPI, Next.js, Crawl4AI, GenAI, AWS*

[GitHub](#) ↗

- Designed an AI lead-discovery pipeline spanning LLM query planning, geo-targeted crawling, contact extraction, semantic search, and predictive scoring.
- Deployed the end-to-end platform on AWS using EC2, RDS, S3, CloudFront, and SES.

ACHIEVEMENTS

- **1st Runner-Up**, I Love Hackathon: Chennai Web3 (**2nd of 49 finalists**, selected from **900+ applicants**).
- **Product Launch:** co-authored Nethermind's *defi-skills* launch post with the marketing team, generating **20K+ views on X**.
- **Open-Source Contributor:** authored PRs and issues and reviewed contributions across **spotDL**, **Formstr**, **Reflex**, **Ansible**, **Open Library (Internet Archive)**, and **Lightning Dev Kit**.

EDUCATION

Vellore Institute of Technology, Chennai

Chennai, India

B.Tech, Computer Science and Engineering, CGPA: 8.03/10

Aug 2023 to May 2027

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, JavaScript, C / C++, C#, Solidity

Systems / Protocols: Bitcoin Lightning, LDK, gRPC, Protobuf, EVM, ABI encoding, multi-chain RPC

Backend / Data: FastAPI, Node.js, GraphQL, PostgreSQL, SQLite, SQL Server, MongoDB

AI / ML: PyTorch, Hugging Face, LoRA / PEFT, DeepSpeed, FSDP, LangChain, semantic search

Infrastructure: AWS, GCP, Docker, Linux, CI/CD, observability and recovery design